

$$\dot{m} = \rho Q = \rho V A$$

$$\sum F = \sum_{out} \dot{m} v - \sum_{in} \dot{m} v$$

$$\dot{W} = F v = T \omega$$

$$\sum T = \sum_{out} \dot{m} (\vec{r} \times \vec{v}) - \sum_{in} \dot{m} (\vec{r} \times \vec{v})$$

$$\dot{W} = \dot{m} g h = \rho g Q h$$

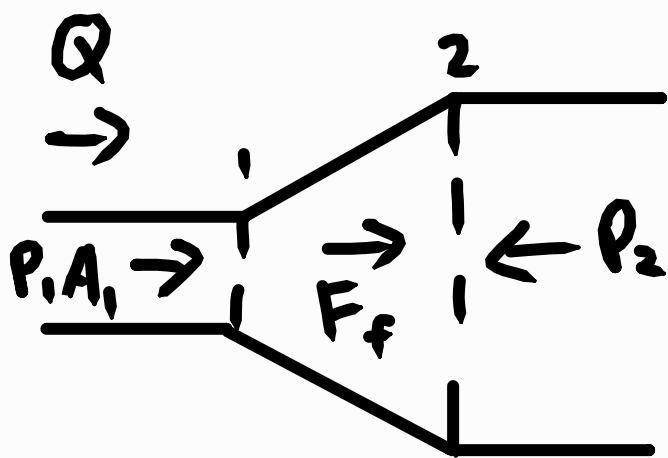
$$v = r \omega$$

$$F = ma$$

$$P = \frac{F}{A}$$

$$= \sum_{out} \dot{m} r v \cos \theta - \sum_{in} \dot{m} r v \cos \theta$$

Linear momentum equation example



$$P_1 A_1 - P_2 A_2 + F_f = \dot{m} (v_2 - v_1)$$

Energy equation:

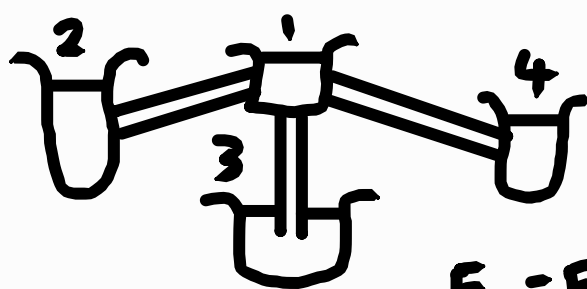
$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + z_1 + h_p - h_L = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + z_2$$

$h_p = h_L = 0 \rightarrow$ Bernoulli's equation

(for smooth laminar flow only)

For multiple pipes:

$\rightarrow$ Use the energy equation to **EACH** one **SEPARATELY!**



Cannot combine:

~~$$E_1 = E_2 + E_3 + E_4$$~~

$$E_1 = E_2 \quad E_1 = E_3 \quad E_1 = E_4 \quad \checkmark$$

Velocity of fluid after flow rate meter:

$$V_2 = \sqrt{\frac{2(P_1 - P_2)}{\rho(1 - \beta^4)}}$$

$$\beta = \frac{d}{D}$$

d = flow rate meter diameter
 D = diameter of pipe **BEFORE** the flow rate meter

Flow rate:

$$Q = C_n Q_{ideal} = C_n A_2 V_2 = C_n \frac{\pi d^2}{4} \sqrt{\frac{2(P_1 - P_2)}{\rho(1 - \beta^4)}}$$

Dimensional analysis

$$Re = \frac{\rho \bar{V} D}{\mu} \quad \nu = \frac{\mu}{\rho}$$

$$Fr = \frac{\bar{V}}{\sqrt{gL}}$$

Flow coefficient:

$$C_Q = \frac{Q}{\omega D^3}$$

Head coefficient:

$$C_H = \frac{gH}{\omega^2 D^2}$$

Power coefficient:

$$C_P = \frac{\dot{W}}{\rho \omega^3 D^5}$$

Common repeating variables:

$$\rho, V, D, \omega$$

Velocity: LT^{-1}

Acceleration: LT^{-2}

Density: ML^{-3}

Force: MLT^{-2}

Pressure: $ML^{-1}T^{-2}$

Viscosity: $ML^{-1}T^{-1}$ 

Torque: ML^2T^{-2}

Power: ML^2T^{-3}

Energy: ML^2T^{-2}

Rotation speed: T^{-1} 

Volumetric flow rate: L^3T^{-1}

Helpful equations:

$$F = ma \quad T = E = Fx \quad P = \frac{F}{A} \quad \dot{W} = Fv$$

Similitude:

1) List everything in a table.

2) Use the dimensionless properties to compare and find the missing values.

<u>Viscous flow</u>	Laminar $Re \leq 2100$	Turbulent $Re > 4000$
Pressure drop	$\frac{\Delta P}{L} = -\frac{2\tau}{r} - \rho g \frac{\Delta z}{L}$	
Velocity profile	$V = V_c \left[1 - \left(\frac{2r}{D}\right)^2\right]$	$V = V_c \left(1 - \frac{2r}{D}\right)^{\frac{1}{n}}$
Average velocity	$\bar{V} = \frac{1}{2} V_c$	$\bar{V} = \frac{2V_c n^2}{(n+1)(2n+1)}$
Centreline velocity	$V_c = \frac{(\Delta P + \rho g \Delta z) D^2}{32 \mu L}$	—
Shear stress	$\tau = \mu \frac{d\bar{V}}{dr} = -\frac{(\Delta P + \rho g \Delta z) r}{2L}$	$\tau = \mu \frac{d\bar{V}}{dr} + \tau_{turb}$

Following the flow direction, point 1 (initial) to point 2 (final):

$$\Delta P = P_1 - P_2 \quad \Delta z = z_1 - z_2$$

Friction factor

Laminar ($Re \leq 2100$)

$$f = \frac{64}{Re}$$

Turbulent ($Re > 4000$)

Smooth pipe, $\epsilon = 0$:

$$f = \frac{0.316}{Re^{0.25}}$$

Rough pipe:

$$\frac{1}{\sqrt{f}} = -2 \log \left(\frac{\frac{\epsilon}{D}}{3.7} + \frac{2.51}{Re \sqrt{f}} \right)$$

If Re is not given, especially for wholly or fully turbulent flow, use the Moody chart's complete turbulence region.

Major loss: $h_L = f \frac{L}{D} \frac{V^2}{2g}$

Minor loss: $h_L = K \frac{V^2}{2g}$

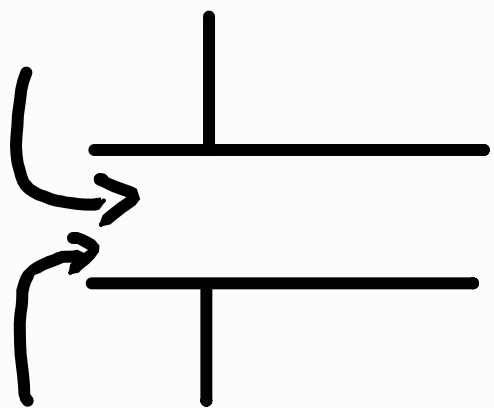
$$\frac{V^2}{2g} = \frac{8Q^2}{g\pi^2 D^4}$$

Exit loss:

$$K = 1$$

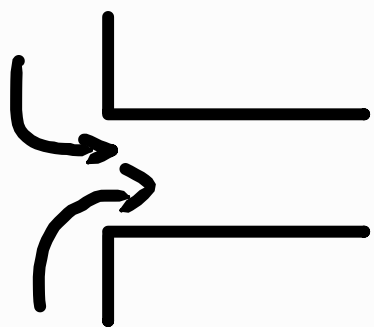
Sudden expansion: $K = \left(1 - \frac{A_{\text{small}}}{A_{\text{big}}}\right)^2 = \left(1 - \frac{D_{\text{small}}^2}{D_{\text{big}}^2}\right)^2$

Entrance loss:



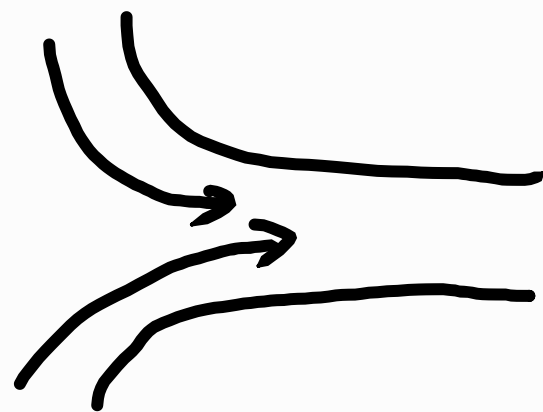
$$K = 0.8$$

Re-entrant



$$K = 0.5$$

Sharp-edged



$$K < 0.5$$

Rounded

Hydraulic diameter: $D_h = \frac{4A}{P}$

$$EGL = \frac{P}{\rho g} + \frac{V^2}{2g} + z$$

$$HGL = \frac{P}{\rho g} + z = EGL - \frac{V^2}{2g}$$

- Pumps and minor losses are vertical lines
- Major loss is a downward sloping straight line
- Gradient of the lines must correspond to the major loss
- The difference between EGL and HGL must correspond to the velocity
- HGL must **increase** when there is **sudden expansion**

Pipes in series:

$$Q = Q_1 = Q_2 = \dots$$

$$h_L = \sum h_{\text{major}} + \sum h_{\text{minor}}$$

Pipes in parallel:

$$Q = Q_1 + Q_2 + \dots$$

$$\begin{aligned} h_1 &= h_{\text{major},1} + \sum h_{\text{minor},1} \\ h_2 &= h_{\text{major},2} + \sum h_{\text{minor},2} \\ &\vdots \end{aligned}$$

Turbomachines

Pumps add energy:

Energy equation + h_p

Turbines extract energy:

Energy equation - h_t

Axial pump: High flow rate, low pressure rise

Centrifugal pump: Moderate flow rate, high pressure rise

$$\dot{W} = \rho g h_p Q$$

$$\eta = \frac{\dot{W}}{\dot{W}_{\text{shaft}}} = \frac{\rho g h_p Q}{\dot{W}_{\text{shaft}}}$$

Pump equation:

$$h_p = \Delta z - A Q^2$$

System characteristic:

$$\cancel{\frac{P_1}{\rho g}} + \cancel{\frac{V_1^2}{2g}} + z_1 + h_s - h_L = \cancel{\frac{P_2}{\rho g}} + \cancel{\frac{V_2^2}{2g}} + z_2$$

Δz : Shutoff head, $Q=0$

$h_p=0$: Free delivery

$$h_s = z_2 - z_1 + h_L$$

$$= z_2 - z_1 + B Q^2$$

Operating point:

$$h_s = h_p$$

Pumps in series:

$$Q = Q_1 = Q_2 = \dots$$

$$h = h_1 + h_2 + \dots$$

Pumps in parallel:

$$Q = Q_1 + Q_2 + \dots$$

$$h = h_1 = h_2 = \dots$$

Cavitation

All pressures must be ABSOLUTE!

Whether a point P in a pipe will cavitate:

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 - h_L = \frac{P_P}{\rho g} + \frac{V_P^2}{2g} + z_P$$

Solve for P_P ,

Cavitation occurs if $P_P < P_v$

$$NPSH = \frac{P_s - P_v}{\rho g} + \frac{V_s^2}{2g}$$

Where s is the pump suction or inlet.

To avoid cavitation:

$$NPSH_A \geq NPSH_R$$

Assume $NPSH_R = 0$ if not given.